Po-Sheng Cheng

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Portfolio: https://bencer3283.github.io/art/

Master of Industrial Design, Rhode Island School of Design, exp. Jun. 2026
Cross-enrolled at School of Engineering, Brown University
B.Sc. Bio-Mechatronics Engineering, National Taiwan University, Jan 2023

Skills

- Electrical: PCB design (Altium), STM32, Raspberry Pi, system interface (I2C, SPI, UART)
- Coding: C/C++, JavaScript (React, Express, MongoDB), Dart (Flutter), Python (PyTorch, NumPy), Git, Linux

Experiences

Faculty of Record, Dept. of Industrial Desing, RISD, Jan. - Feb. 2026 — Providence, RI

- Taught a introductory studio class (ID1531) in physical computing and embedded system development. Employed theoretical lectures, demo, and hands-on workshop to cover topic like microcontroller, circuit theory, power rails, binary arithmetic, serial communication, HTTP, and machine learning.
- Blended technical skill with discussion on the societal impact of technology with framework like Science and Technology Studies.

Design Technologist Intern, Smart Design, Jun. - Aug. 2025 — Brooklyn, NY

- Developed embedded firmware (STM32) for prototyping the HMI of a household product, while working with UX design to test and fine-tune user interaction flow and hardware user interface.
- Developed testing units for motor and pneumatic mechanisms in the product, including the electrical hardware, embedded firmware and host software. (PyQt)

Product Development Apprentice, Google, Jun. - Sep. 2024 — Taipei, Taiwan

- Led a team of 5 to design a Hardware-as-a-Service platform for managing lost items in public facilities. Laid out the system architecture of the hardware kiosk, user-facing app and web services.
- My prototypes of Raspberry Pi and ESP32 hardware, a React-based website, a mobile app and a RESTful API service drove the development direction from a single hardware product to an integrated platform.

Electro-mechanical Engineering Intern, Logitech, Feb. - Jun. 2022 — Hsinchu, Taiwan

- Created a brand new breed of keyboard technology by conducting quantitative user research with 50 interviewees and innovating an Arduino microcontroller-based prototype to control actuators like stepper motor and electromagnet. Also lots of CAD (Creo) and SLA/FDM prototyping on the mechanical design.
- The prototype recieved widespread praise across many departments (pm, engineering, design) in the company while I worked with them to support guiding the direction of product development.

Projects

Objectied AI

- Explored several physical embodiment of AI as tangible user interfaces of LLM.
- Prototyped a multi-axis slider interface using hall-effect sensors. Implemented with precise integration of custom PCBA and electromagnetic machanism.

Publications

TeleSHift: Telexisting TUI for Physical Collaboration & Interaction

- Utilized ESP32 to implement a shape-transforming device with a 3D tangible user interface that enabled this work to be awarded **Best Demo** at 2022 ACM Ubicomp conference.
- Designed the robust interactive mechanisms (SolidWorks). My design for manufacturability (DFM) improvements also slashed assembling time by 80% during demo.